

Logical Agents

Chapter 7

- Knowledge-based

Outline

Knowledge-based agents

- Knowledge is the facts and principles accumulated by human beings.
- A k- based agent is composed of a knowledge base and an inference mechanism to infer new sentences and use these sentences to decide what actions to take.
- Example:- if it is raining

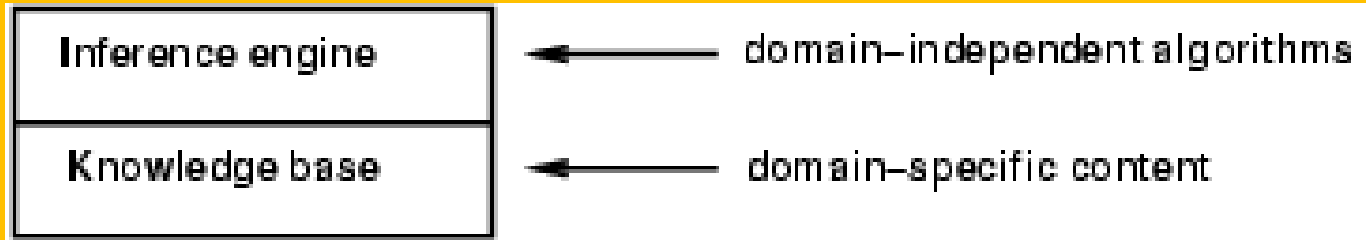
then take an umbrella

weather(raining) $\longrightarrow$ take(umbrella)

premise/antecedent

conclusion/consequent

Knowledge bases



- A Knowledge base is a set of **sentences** in a **formal** language.
- A KB consists of
 - **FACTS** : Statements of truth in the application domain.
 - **PROCEDURAL RULES** : well defined, invariant rules
 - **HEURISTIC** : procedures to be followed when invariant procedurals are not available.

Example :- Gandhian family relation

- FACTS:

- Male(Rajiv)
- Female(Sonia)
- Male(Rahul)
- Female(Priyanka)
- Father(Rajiv, Rahul)
- Mother(Sonia, Rahul)

- RULES:

- Husband(x,y) $\longrightarrow$ male(x)
- Wife(x,y) $\longrightarrow$ female(x)

- INFERENCING:

- Husband(Rajiv , Sonia) $\longrightarrow$ wife(Sonia, Rajiv)

Knowledge bases

- **Declarative** approach to building an agent (or other system):
 - Tell it what it needs to know
- Then it can Ask itself what to do - answers should follow from the KB
- Agents can be viewed at the **knowledge level**
i.e., what they know, regardless of how implemented
- Or at the **implementation level**
 - i.e., data structures in KB and algorithms that manipulate them

A simple knowledge-based agent

```
function KB-AGENT(percept) returns an action  
  static: KB, a knowledge base  
           t, a counter, initially 0, indicating time  
  
  TELL(KB, MAKE-PERCEPT-SENTENCE(percept, t))  
  action  $\leftarrow$  ASK(KB, MAKE-ACTION-QUERY(t))  
  TELL(KB, MAKE-ACTION-SENTENCE(action, t))  
  t  $\leftarrow$  t + 1  
  return action
```

- The agent must be able to.
 - Represent states, actions, etc.
 - Incorporate new percepts

Wumpus World PEAS description

- Performance measure

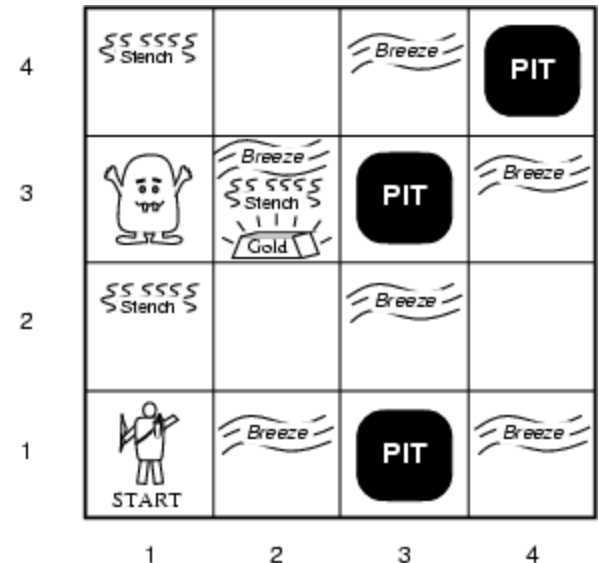
- gold +1000, death -1000
- -1 per step, -10 for using the arrow

- Environment

- Squares adjacent to wumpus are smelly
- Squares adjacent to pit are breezy
- Glitter iff gold is in the same square
- Shooting kills wumpus if you are facing it
- Shooting uses up the only arrow
- Grabbing picks up gold if in same square
- Releasing drops the gold in same square

- Sensors: Stench, Breeze, Glitter, **Bump**, **Scream**

- Actuators: Left turn, Right turn, Forward, Grab, Release, Shoot



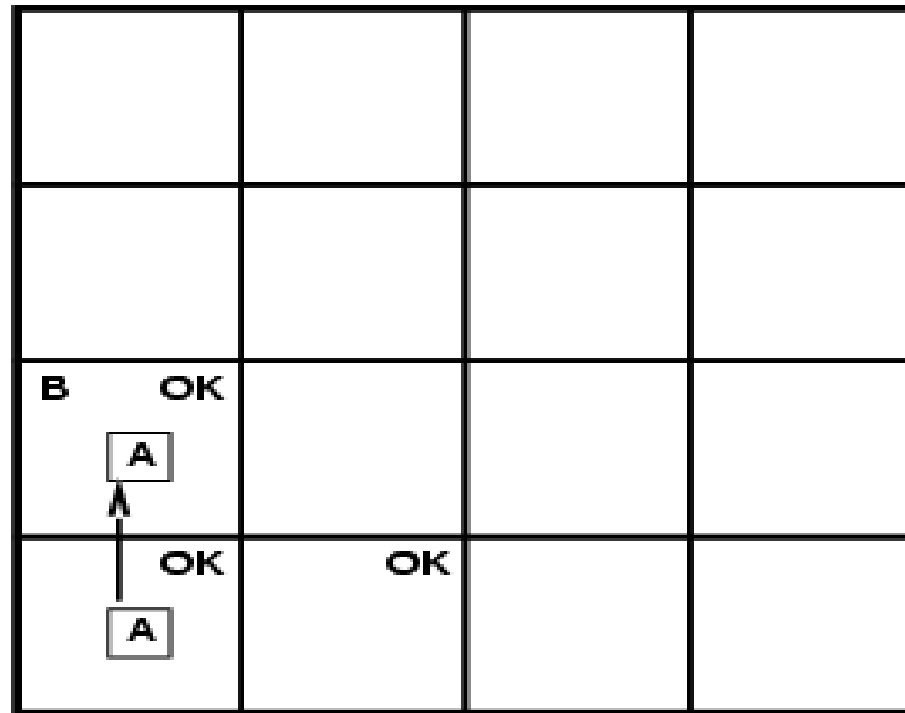
Wumpus world characterization

- Fully_Observable No – only **local** perception
- Deterministic Yes – outcomes exactly specified
- Episodic No – sequential at the level of actions
- Static Yes – Wumpus and Pits do not move
- Discrete Yes
- Single-agent? Yes – Wumpus is essentially a natural feature

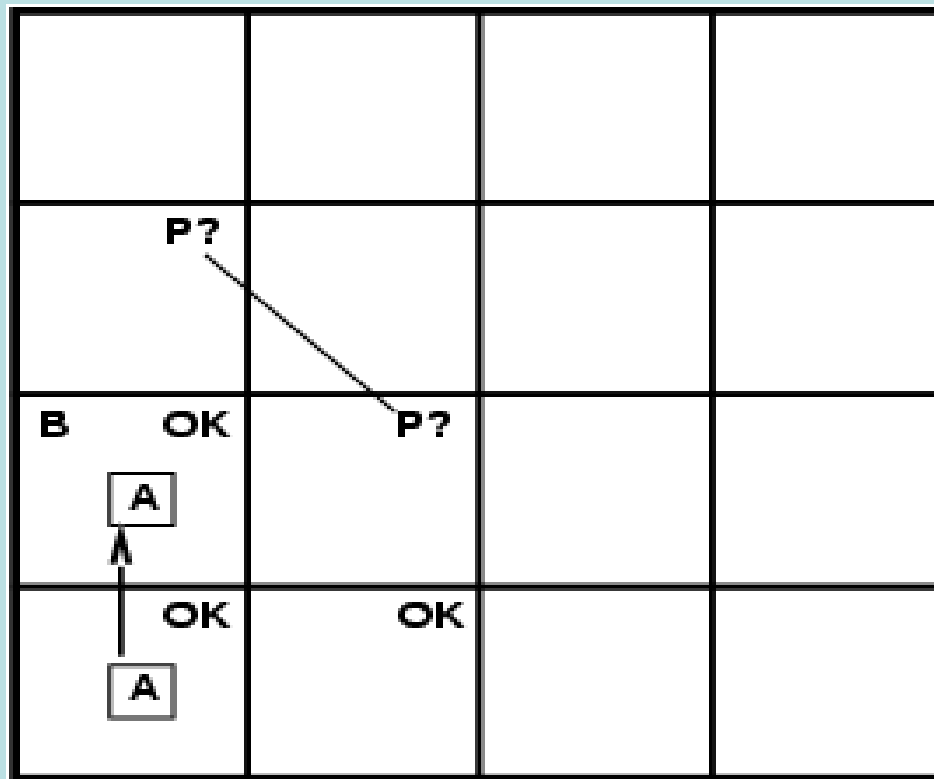
Exploring a wumpus world

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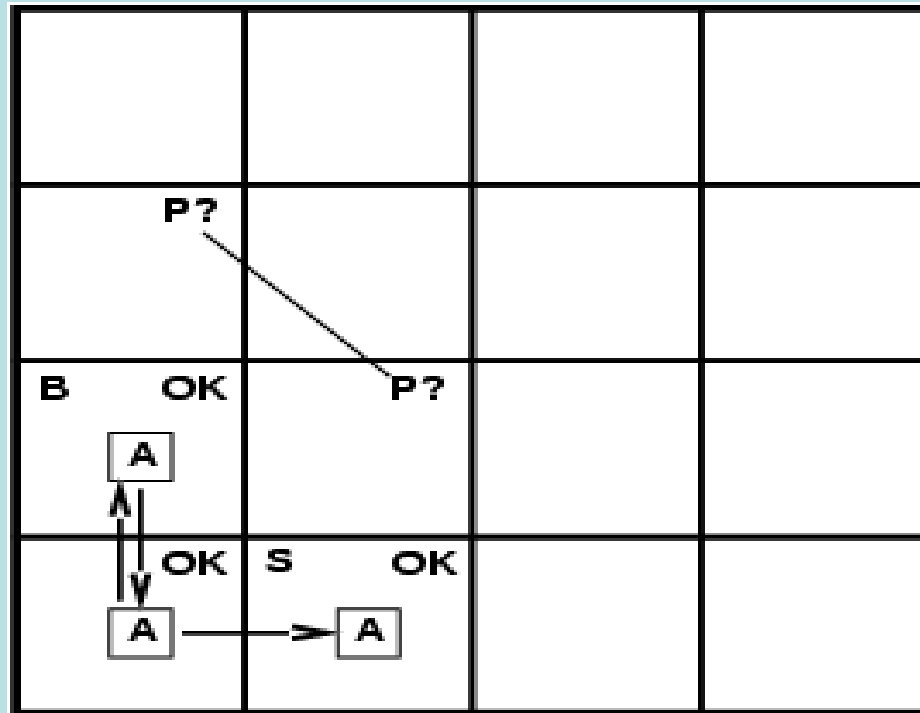
Exploring a wumpus world



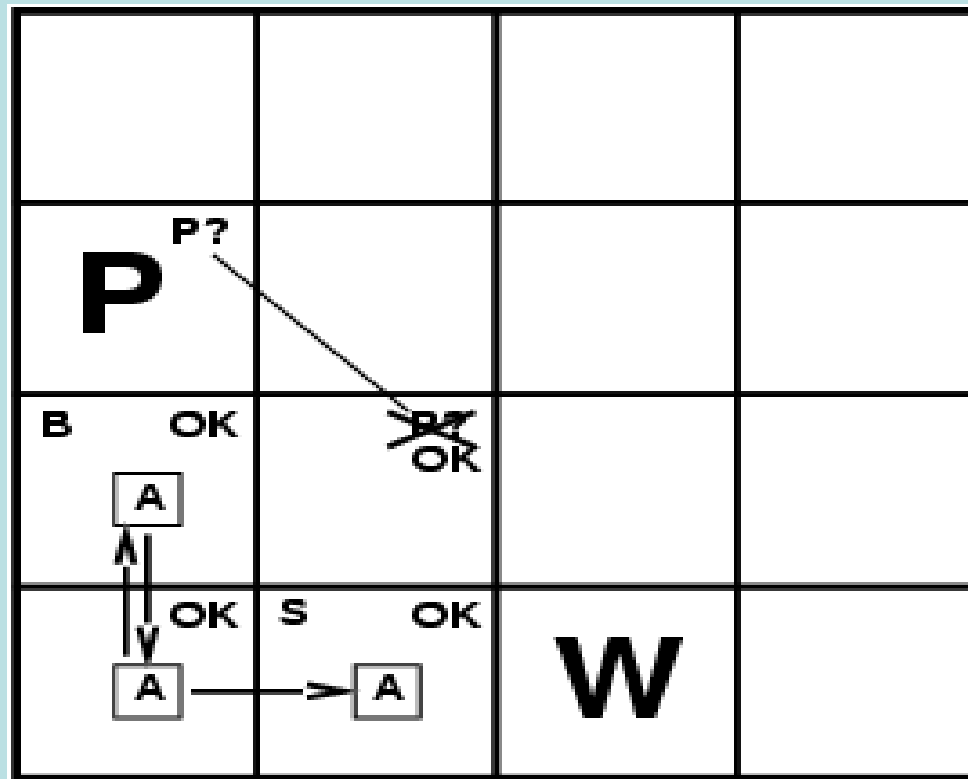
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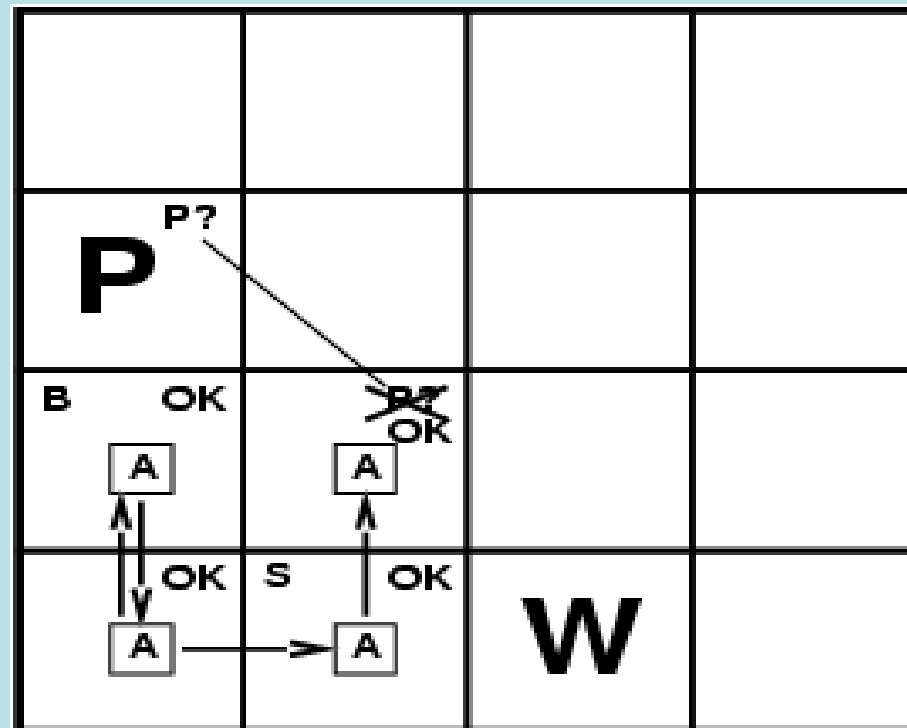
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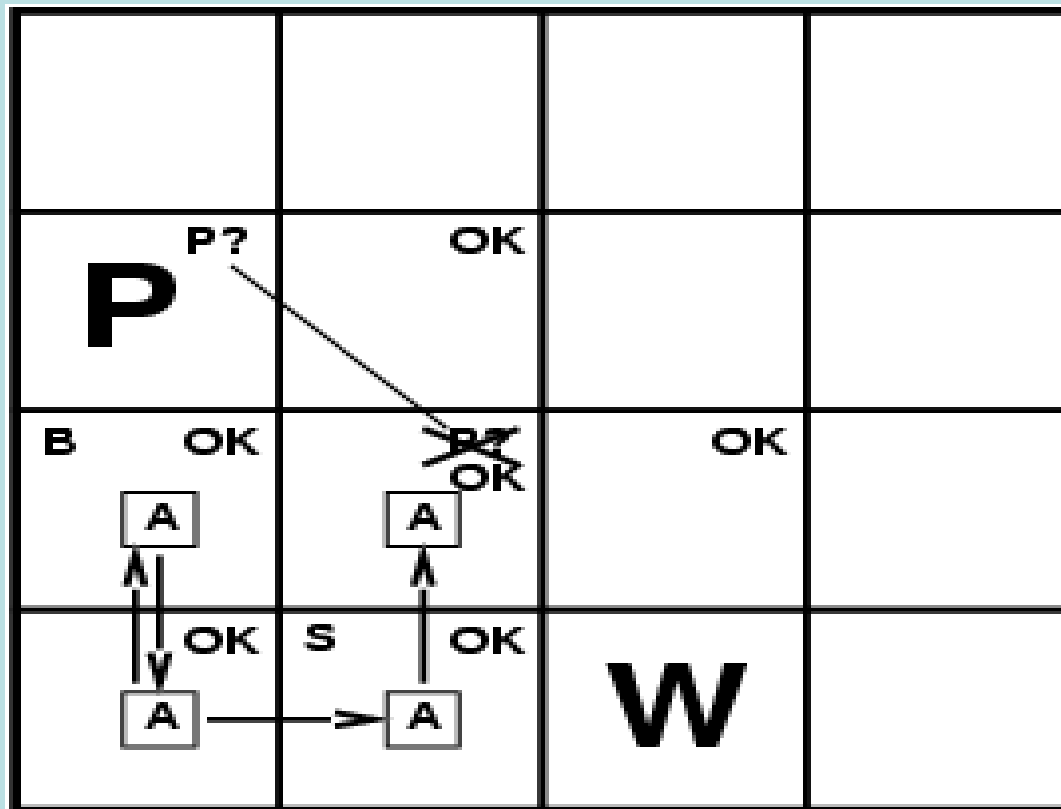
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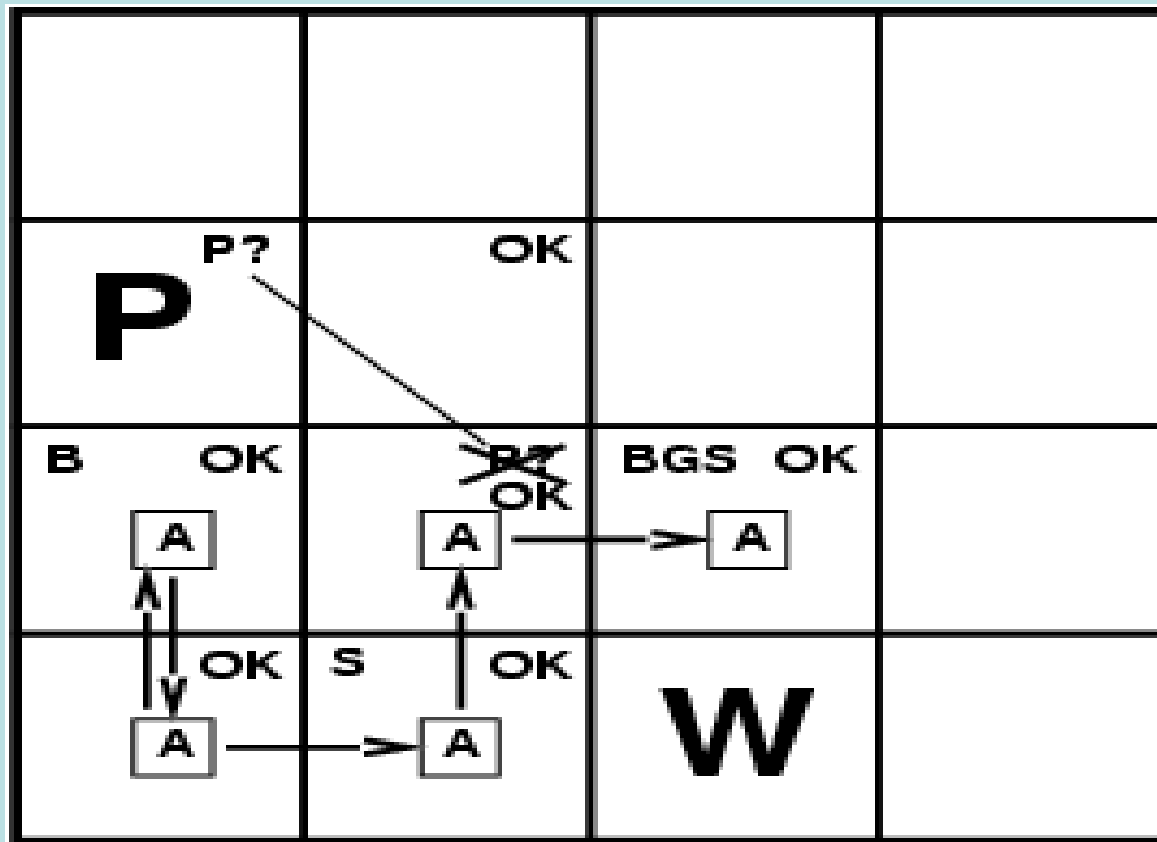
Exploring a wumpus world



Exploring a wumpus world



Exploring a wumpus world



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LK- Representation using symbolic logic

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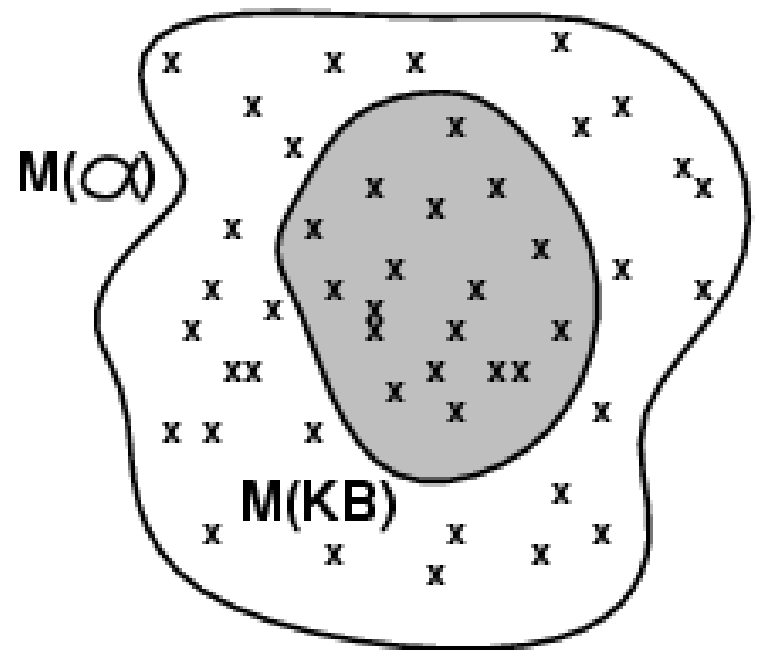
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Logic in general

Entailment

Entailment means that

Models

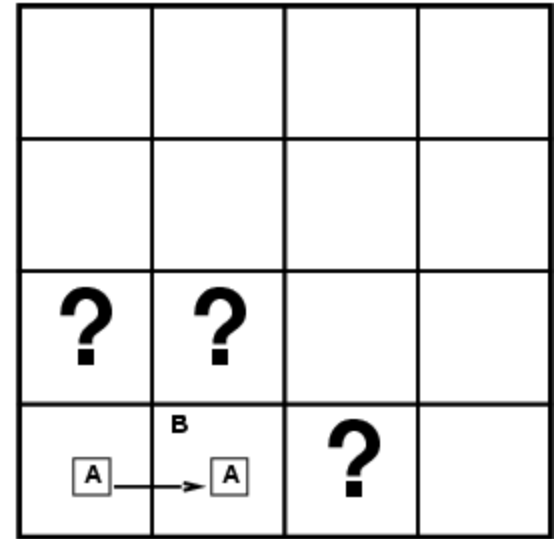


Entailment in the wumpus world

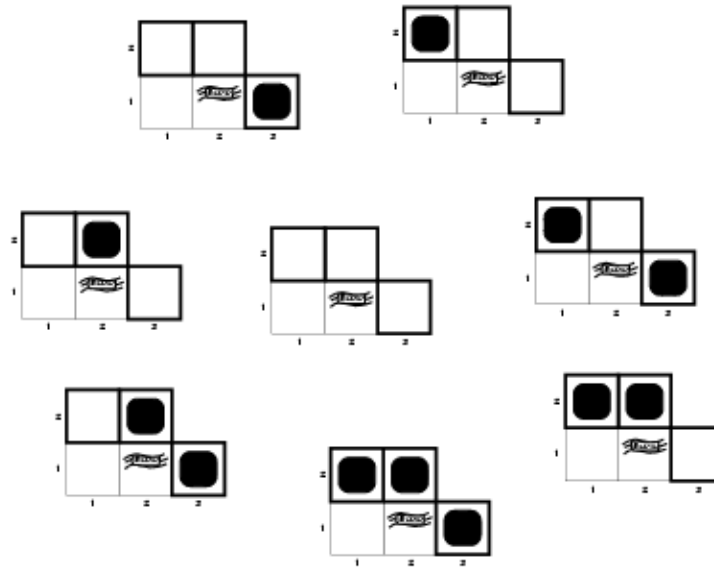
Situation after detecting
nothing in [1,1], moving
right, breeze in [2,1]

Consider possible models for
KB assuming only pits

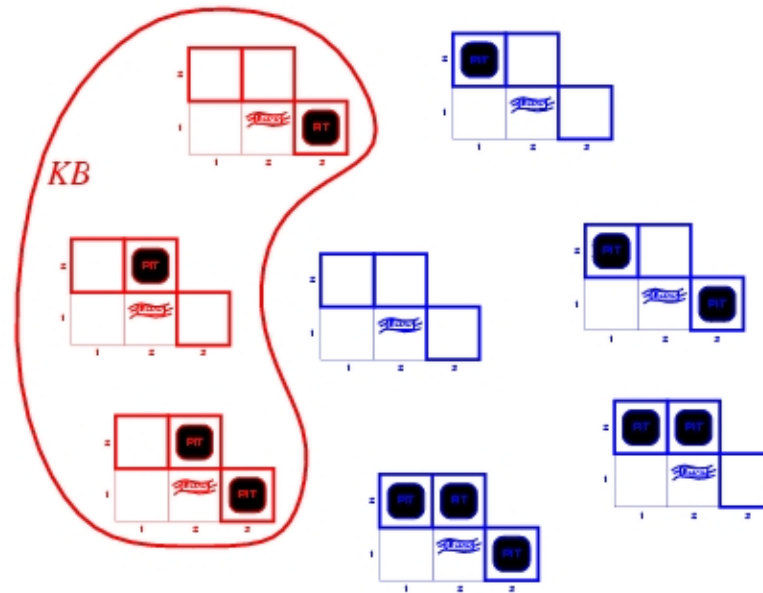
3 Boolean choices $\Rightarrow$ 8
possible models



Wumpus models

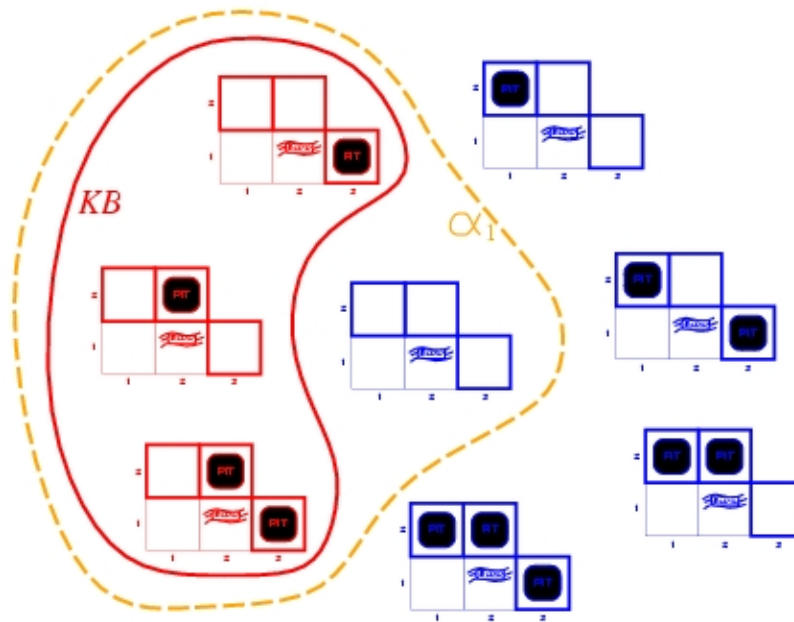


Wumpus models



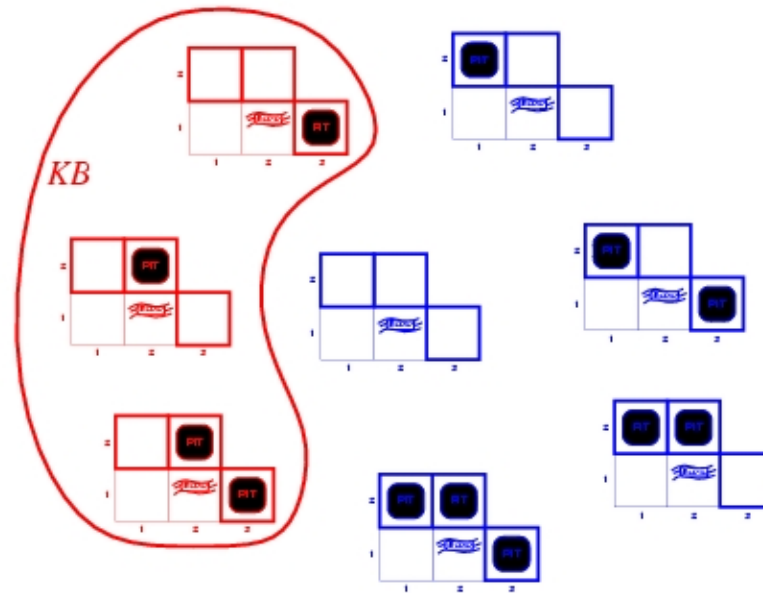
- KB = wumpus-world rules + observations

Wumpus models



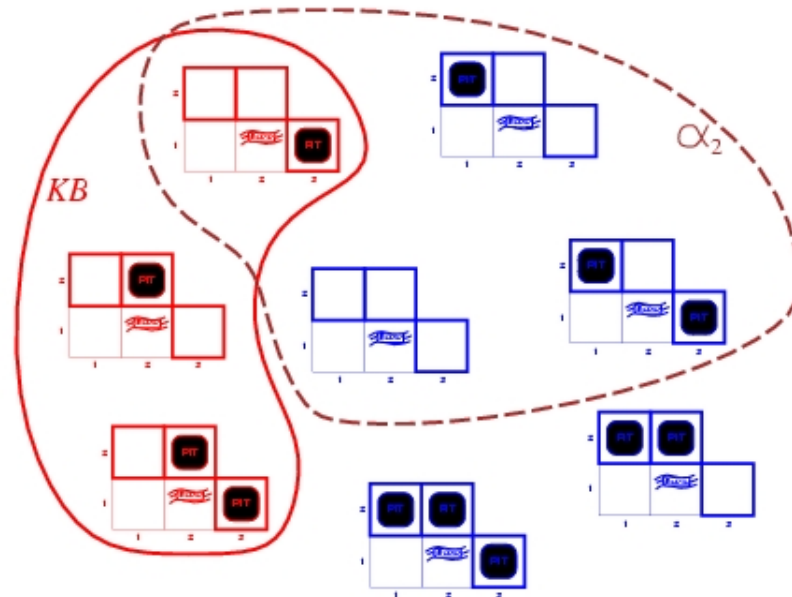
- KB = wumpus-world rules + observations
- α_1 = "[1,2] is safe", $KB \models \alpha_1$, proved by model checking

Wumpus models



- KB = wumpus-world rules + observations

Wumpus models



- KB = wumpus-world rules + observations
- α_2 = "[2,2] is safe", $KB \not\models \alpha_2$

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Inference

Propositional logic: Syntax

Propositional logic

Propositional logic: Semantics

Each model specifies true/false for each proposition symbol

E.g. $P_{1,2}$	$P_{2,2}$	$P_{3,1}$	
	false	true	false

With these symbols, 8 possible models, can be enumerated automatically.

Rules for evaluating truth with respect to a model m :

$\neg S$	is true iff	S is false	
$S_1 \wedge S_2$	is true iff	S_1 is true and	S_2 is true
$S_1 \vee S_2$	is true iff	S_1 is true or	S_2 is true
$S_1 \Rightarrow S_2$	is true iff	S_1 is false or	S_2 is true
i.e.,	is false iff	S_1 is true and	S_2 is false
$S_1 \Leftrightarrow S_2$	is true iff	$S_1 \Rightarrow S_2$ is true and	$S_2 \Rightarrow S_1$ is true

Simple recursive process evaluates an arbitrary sentence, e.g.,

$$\neg P_{1,2} \wedge (P_{2,2} \vee P_{3,1}) = \text{true} \wedge (\text{true} \vee \text{false}) = \text{true} \wedge \text{true} = \text{true}$$

Truth tables for connectives

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>true</i>
<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>
<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>
<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>

Wumpus world sentences

Let $P_{i,j}$ be true if there is a pit in $[i, j]$.

Let $B_{i,j}$ be true if there is a breeze in $[i, j]$.

$$\neg P_{1,1}$$

$$\neg B_{1,1}$$

$$B_{2,1}$$

- "Pits cause breezes in adjacent squares"

$$B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$$

$$B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$$

Logical equivalence

- Two sentences are **logically equivalent** iff true in same models: $\alpha \equiv \beta$ iff $\alpha \models \beta$ and $\beta \models \alpha$

$$\begin{aligned}(\alpha \wedge \beta) &\equiv (\beta \wedge \alpha) && \text{commutativity of } \wedge \\(\alpha \vee \beta) &\equiv (\beta \vee \alpha) && \text{commutativity of } \vee \\((\alpha \wedge \beta) \wedge \gamma) &\equiv (\alpha \wedge (\beta \wedge \gamma)) && \text{associativity of } \wedge \\((\alpha \vee \beta) \vee \gamma) &\equiv (\alpha \vee (\beta \vee \gamma)) && \text{associativity of } \vee \\\neg(\neg\alpha) &\equiv \alpha && \text{double-negation elimination} \\(\alpha \Rightarrow \beta) &\equiv (\neg\beta \Rightarrow \neg\alpha) && \text{contraposition} \\(\alpha \Rightarrow \beta) &\equiv (\neg\alpha \vee \beta) && \text{implication elimination} \\(\alpha \Leftrightarrow \beta) &\equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) && \text{biconditional elimination} \\\neg(\alpha \wedge \beta) &\equiv (\neg\alpha \vee \neg\beta) && \text{de Morgan} \\\neg(\alpha \vee \beta) &\equiv (\neg\alpha \wedge \neg\beta) && \text{de Morgan} \\(\alpha \wedge (\beta \vee \gamma)) &\equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) && \text{distributivity of } \wedge \text{ over } \vee \\(\alpha \vee (\beta \wedge \gamma)) &\equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) && \text{distributivity of } \vee \text{ over } \wedge\end{aligned}$$

Validity and satisfiability

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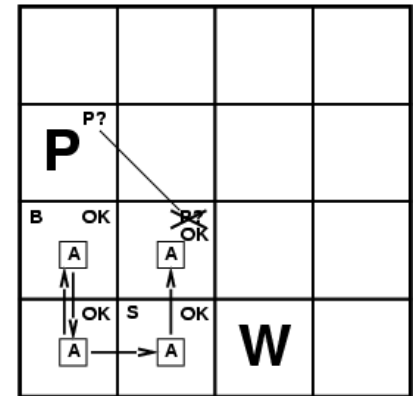
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Proof methods

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Resolution



Conversion to CNF

$$B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$$

1. Eliminate $\Leftrightarrow$, replacing $\alpha \Leftrightarrow \beta$ with $(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)$. $(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$ Eliminate $\Rightarrow$, replacing $\alpha \Rightarrow \beta$ with $\neg\alpha \vee \beta$. $(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge (\neg(P_{1,2} \vee P_{2,1}) \vee B_{1,1})$

5. Move $\neg$ inwards using de Morgan's rules and double-negation:

$$(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge ((\neg P_{1,2} \wedge \neg P_{2,1}) \vee B_{1,1})$$

6. Apply distributive law ($\wedge$ over $\vee$) and flatten: $\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge (\neg P_{1,2} \vee B_{1,1}) \wedge (\neg P_{2,1} \vee B_{1,1})$ (

Now it is in CNF form.

Resolution rules

- Modus Ponens

$$\frac{\alpha \implies \beta, \alpha}{\beta}$$

- And-Elimination

$$\frac{\alpha \wedge \beta}{\alpha}$$

- Elimination of complementary literals

$$\frac{\alpha \vee \beta, \sim \alpha \vee \gamma}{\beta \vee \gamma}$$

Resolution algorithm

- Proof by contradiction, i.e., show $KB \wedge \neg\alpha$ un-satisfiable

```
function PL-RESOLUTION( $KB, \alpha$ ) returns true or false  
   $clauses \leftarrow$  the set of clauses in the CNF representation of  $KB \wedge \neg\alpha$   
   $new \leftarrow \{ \}$   
  loop do  
    for each  $C_i, C_j$  in  $clauses$  do  
       $resolvents \leftarrow$  PL-RESOLVE( $C_i, C_j$ )  
      if  $resolvents$  contains the empty clause then return true  
       $new \leftarrow new \cup resolvents$   
    if  $new \subseteq clauses$  then return false  
     $clauses \leftarrow clauses \cup new$ 
```

Resolution example

Q1. Prove that there is no pit in [1, 2].

Ans:-

From the KB we have:-

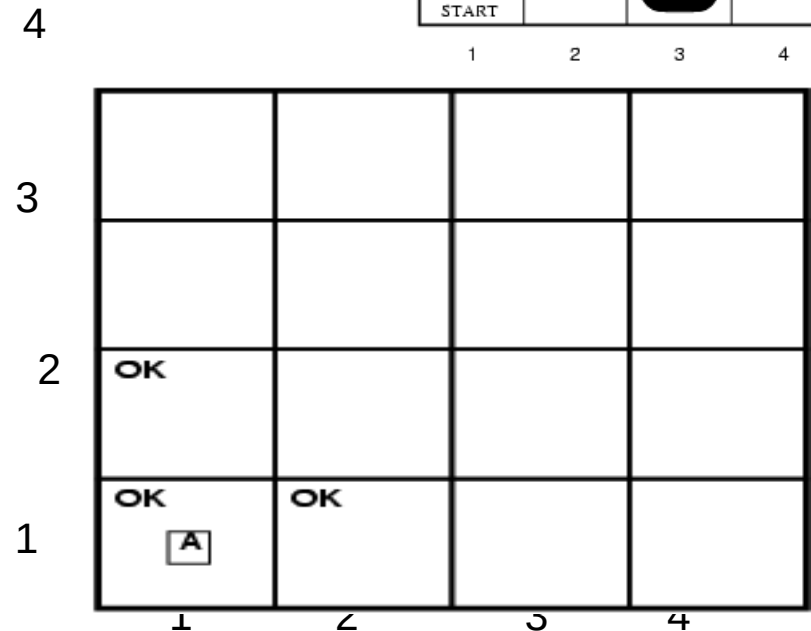
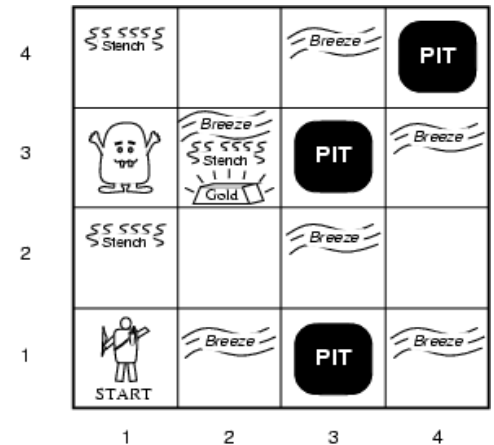
from observation:

R1: $\sim P_{1,1}$,

R2: $\sim B_{1,1}$

From rule:-

R3. $(B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1}))$



Resolution example

Applying bi-conditional elimination to R3:-

$$R4. (B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$$

Applying **And Elimination** on R4

$$R5. ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$$

Applying contra-positive law to R5 :-

$$R6: \sim B_{1,1} \Rightarrow \sim (P_{1,2} \vee P_{2,1})$$

Applying modus ponence to R6 and R2

$$R7: \sim (P_{1,2} \vee P_{2,1})$$

Applying D'morgan's rule to R7:

$$R8: \sim P_{1,2} \wedge \sim P_{2,1}. \text{ applying And elimination to R8:}$$

$$R9: \sim P_{1,2} \text{ and } \sim P_{2,1} \text{ are true.}$$

So, there is no pit in either [1,2] or [2,1]

Resolution example

Q2. Prove that there is no pit in [2, 2] and no pit in [3,1].

Ans:-

From the KB we have:-

from observation:

R10: $B_{1,2}$,

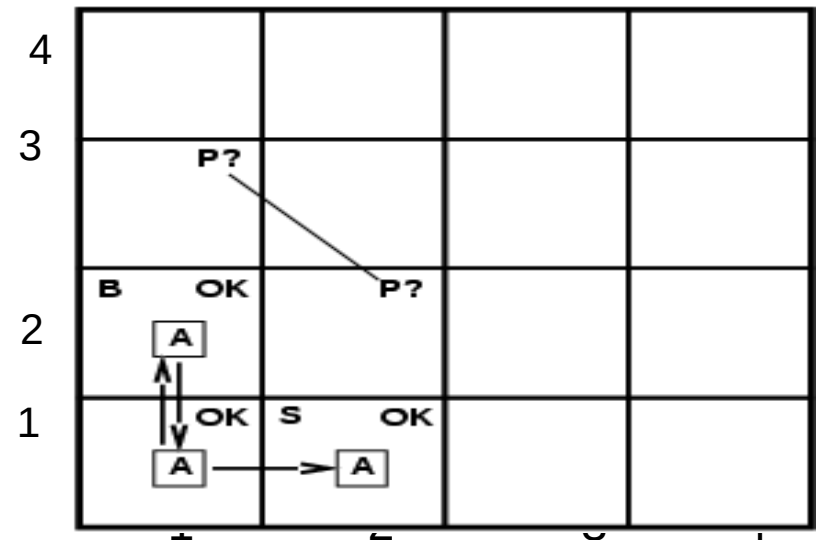
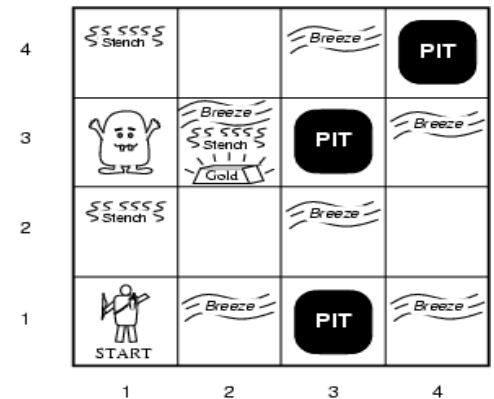
R11: $S_{2,1}$

R12: $\sim B_{2,1}$

From rule:-

R13: $(B_{1,2} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{1,3}))$

R14. $(B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1}))$



Resolution example

Applying bi-conditional elimination to R14:-

R15. $(B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})) \wedge ((P_{1,1} \vee P_{2,2} \vee P_{3,1}) \Rightarrow B_{2,1})$

Applying And elimination on R15

R16. $((P_{1,1} \vee P_{2,2} \vee P_{3,1}) \Rightarrow B_{2,1})$

Applying contra-positive law to R16 :-

R17: $\sim B_{2,1} \Rightarrow \sim (P_{1,1} \vee P_{2,2} \vee P_{3,1})$

Applying modus ponence to R17 and R12

R18: $\sim (P_{1,1} \vee P_{2,2} \vee P_{3,1})$

Applying D'morgan's rule to R18:

R19: $\sim P_{1,1} \wedge \sim P_{2,2} \wedge \sim P_{3,1}$

Applying And elimination to R19:

R20: $\sim P_{1,1}$ and $\sim P_{2,2}$ and $\sim P_{3,1}$ are true.

So, there is no pit in [1,1] , [2,2] or [3,1]

Resolution example

Q3. Prove that there is Wumpus in $[3, 1]$

Ans:- from observation:-

R1: $\sim S_{1,2}$

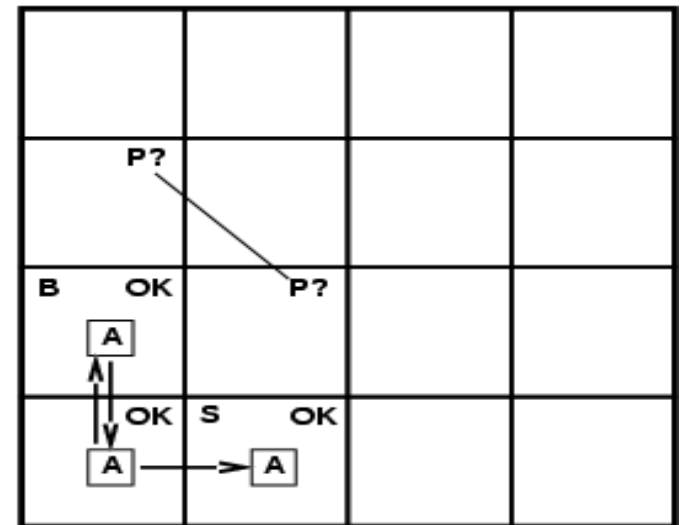
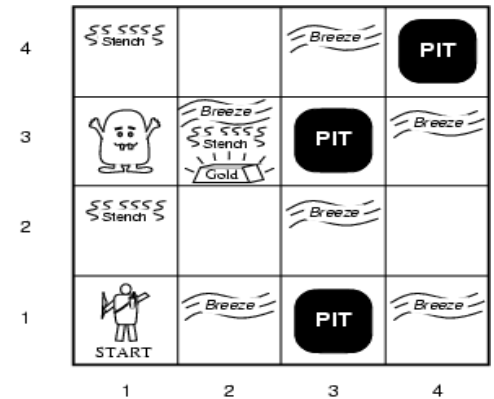
R2: $S_{2,1}$

R3: $\sim W_{1,1}$, $\sim W_{1,2}$, $\sim W_{2,1}$

From rule:-

$$\text{R4: } S_{1,2} \Leftrightarrow (W_{1,1} \vee W_{2,2} \vee W_{1,3})$$

R5: $S_{2,1} \Leftrightarrow (W_{1,1} \vee W_{2,2} \vee W_{3,1})$



Resolution example

Applying bi-conditional elimination to R4:-

$$R6: S_{1,2} \Rightarrow (W_{1,1} \vee W_{2,2} \vee W_{1,3}) \wedge ((W_{1,1} \vee W_{2,2} \vee W_{1,3}) \Rightarrow S_{1,2})$$

Applying end elimination on R6

$$R7: ((W_{1,1} \vee W_{2,2} \vee W_{1,3}) \Rightarrow S_{1,2})$$

Applying contra-positive law to R7 :-

$$R8: \sim S_{1,2} \Rightarrow \sim (W_{1,1} \vee W_{2,2} \vee W_{1,3})$$

Applying modus ponence to R1 and R8

$$R9: \sim (W_{1,1} \vee W_{2,2} \vee W_{1,3})$$

Applying D'morgan's rule to R9:

$$R10: \sim W_{1,1} \wedge \sim W_{2,2} \wedge \sim W_{1,3}$$

Applying And elimination to R10:

$$R11: \sim W_{1,1} \text{ and } \sim W_{2,2} \text{ and } \sim W_{1,3} \text{ are true.}$$

Resolution example

Applying bi-conditional elimination to R5:-

$$R12: (S_{2,1} \Rightarrow (W_{1,1} \vee W_{2,2} \vee W_{3,1})) \wedge ((W_{1,1} \vee W_{2,2} \vee W_{3,1}) \Rightarrow S_{2,1})$$

Applying end elimination on R12

$$R13. A (S_{2,1} \Rightarrow (W_{1,1} \vee W_{2,2} \vee W_{3,1}))$$

Applying modus ponence to R2 and R13

$$R14: (W_{1,1} \vee W_{2,2} \vee W_{3,1})$$

Applying end elimination to R14 and $\sim W_{1,1}$:

$$R15: (W_{2,2} \vee W_{3,1})$$

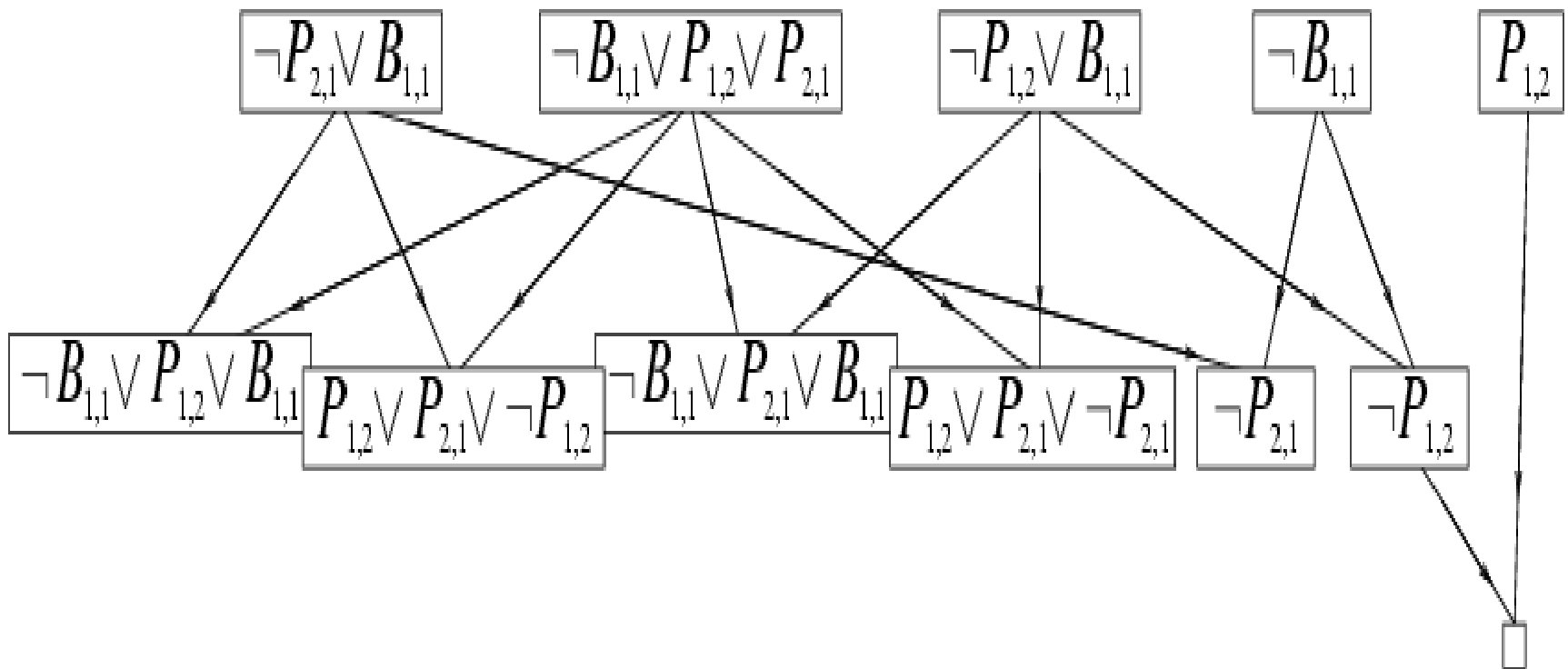
Applying end elimination to R15 and $\sim W_{2,2}$:

$$R15: W_{3,1}$$

So there is wumpus in [3,1].

Resolution by contradiction example

- Convert sentences of KB into CNF form.
- Consider $(KB \wedge \neg\alpha)$
- A nil proves that α is true.



Forward and backward chaining

Forward and backward reasoning/chaining

• Forward chaining:

– I

Forward chaining

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

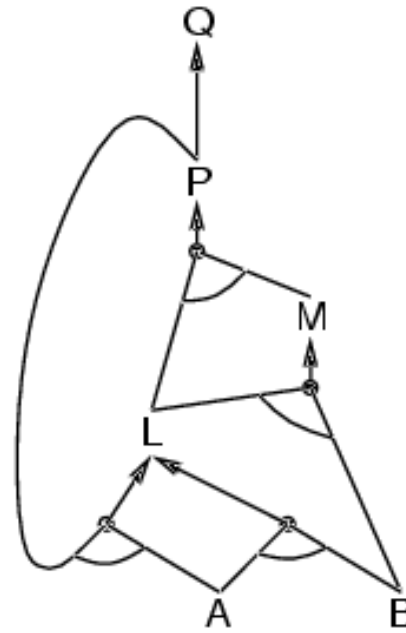
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

B



Forward chaining algorithm

function PL-FC-ENTAILS?(*KB, q*) **returns** *true* or *false*

local variables: *count*, a table, indexed by clause, initially the number of premises
inferred, a table, indexed by symbol, each entry initially *false*
agenda, a list of symbols, initially the symbols known to be true

while *agenda* is not empty **do**

$p \leftarrow \text{POP}(\textit{agenda})$

unless *inferred*[*p*] **do**

inferred[*p*] $\leftarrow \textit{true}$

for each Horn clause *c* in whose premise *p* appears **do**

 decrement *count*[*c*]

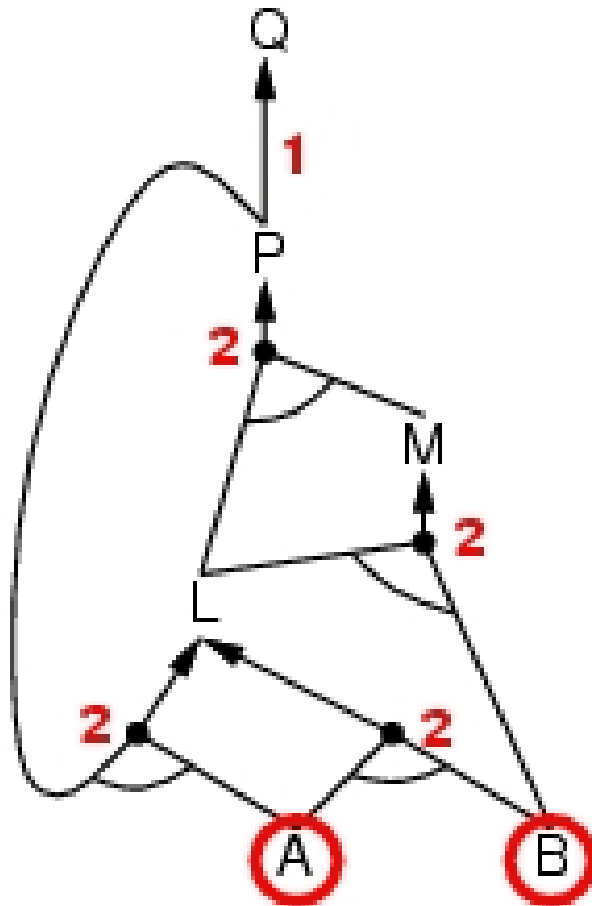
if *count*[*c*] = 0 **then do**

if HEAD[*c*] = *q* **then return** *true*

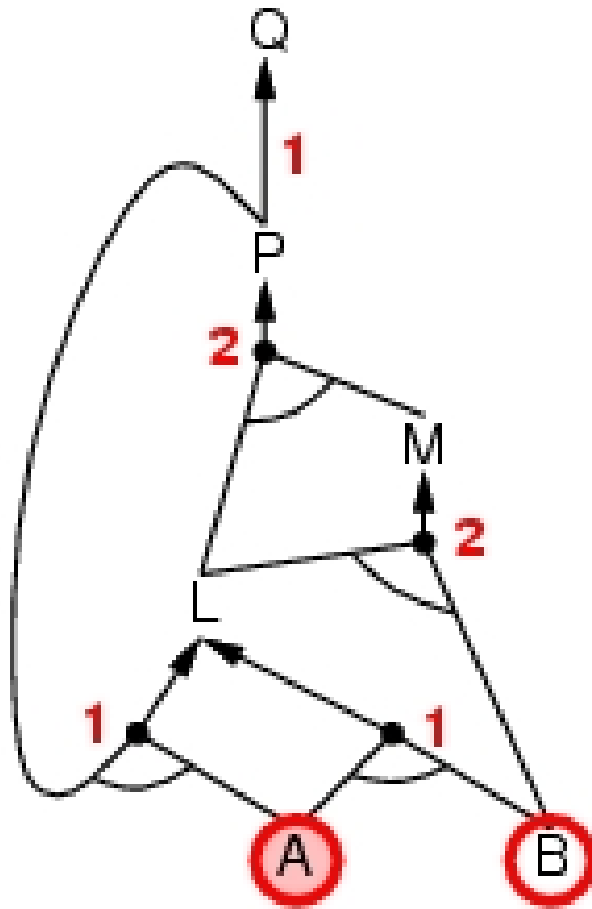
 PUSH(HEAD[*c*], *agenda*)

return *false*

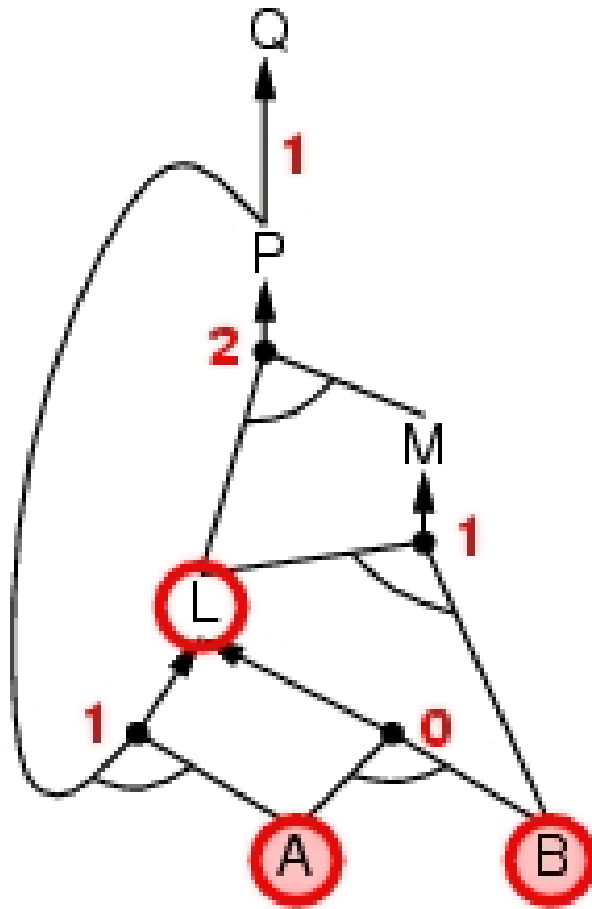
Forward chaining example



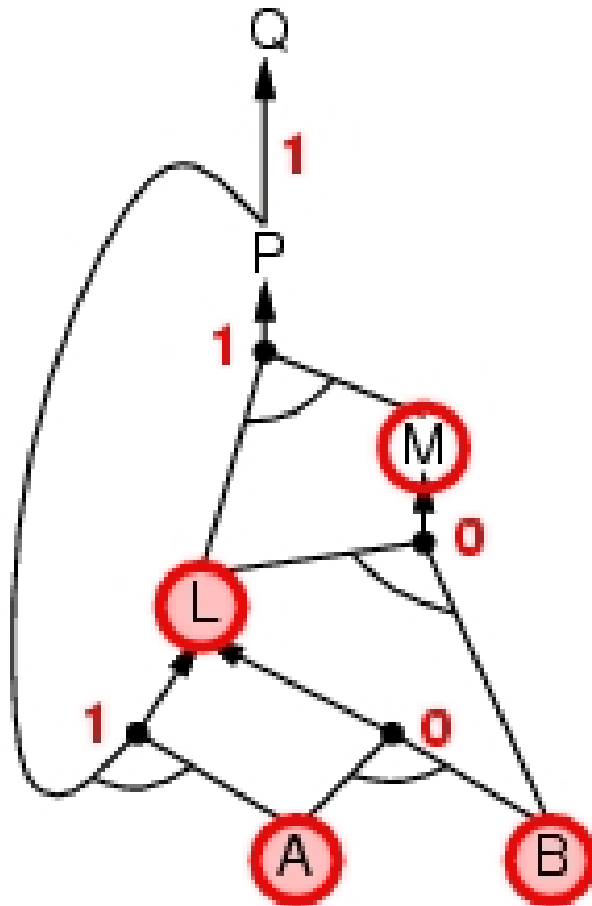
Forward chaining example



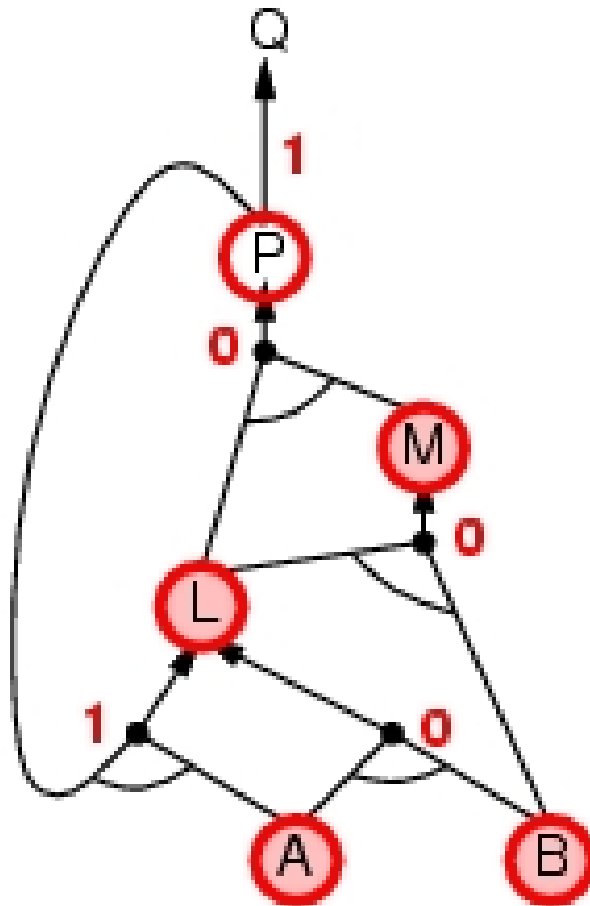
Forward chaining example



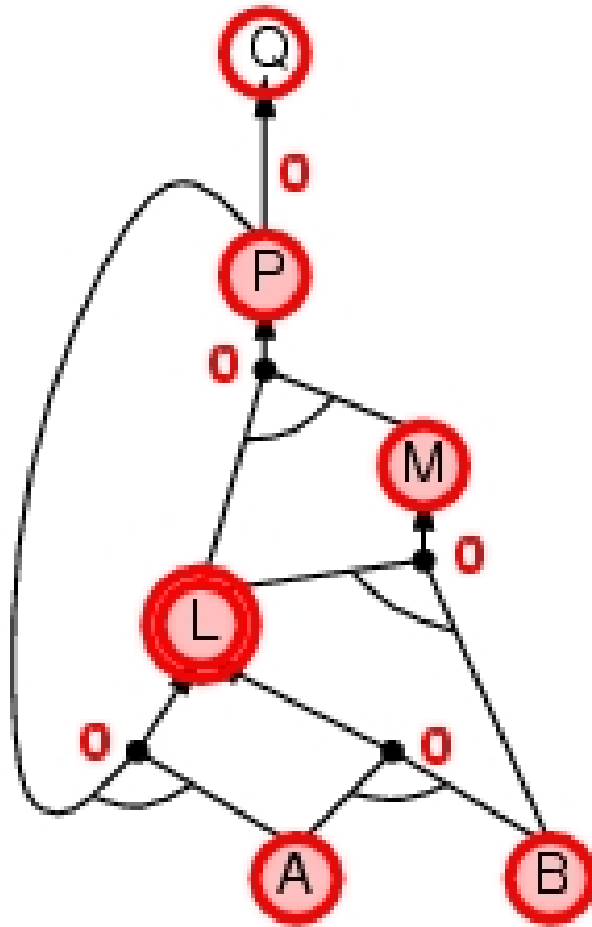
Forward chaining example



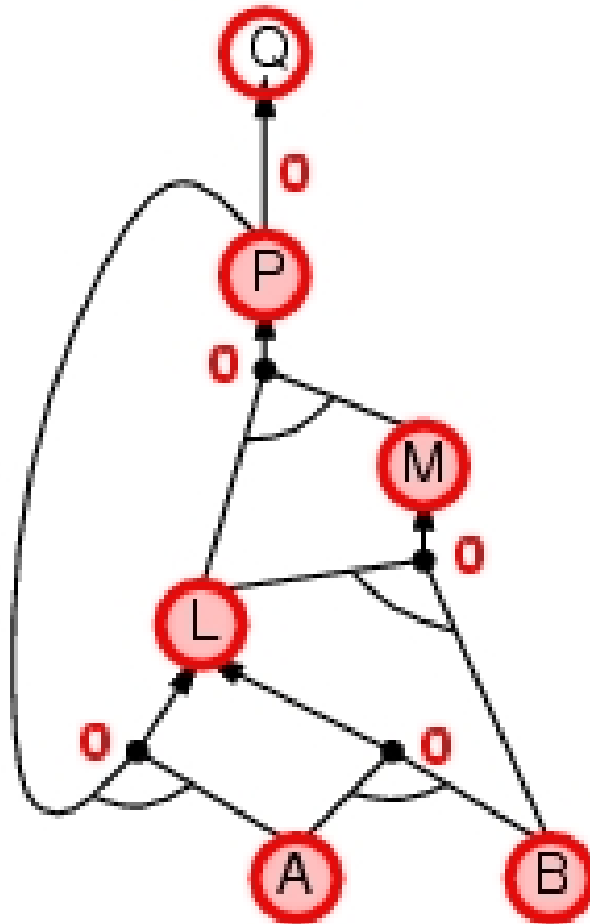
Forward chaining example



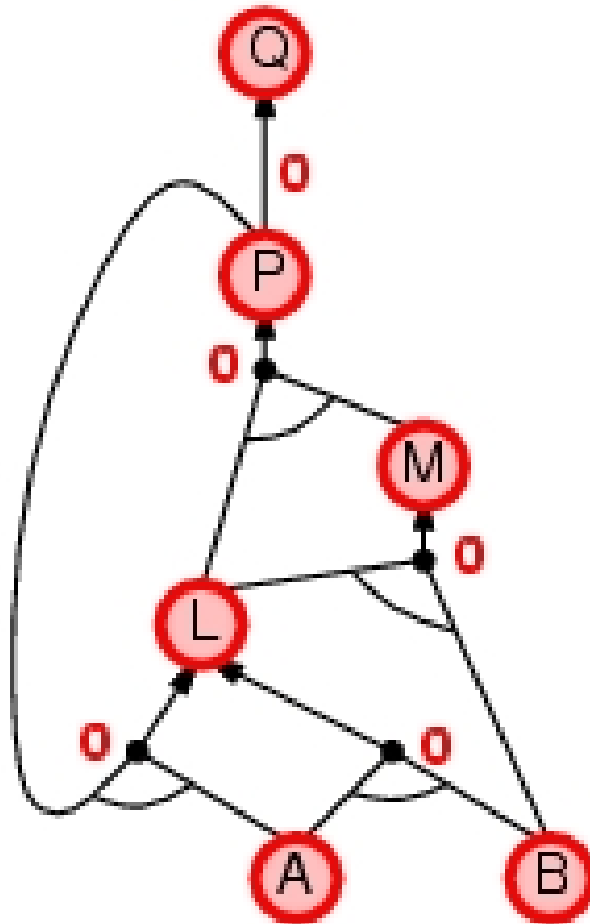
Forward chaining example



Forward chaining example



Forward chaining example



Proof of completeness

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Backward chaining:-

- **It is the process of selection rules by matching the RHS**
-
- **Reasoning begins from the goal state.**
- **The LHS of a selected rule becomes a sub goal which in turn causes another sub goal to be achieved and so on until facts are found to match the lowest sub goal.**
- **This process is also called “goal-driven” inference.**

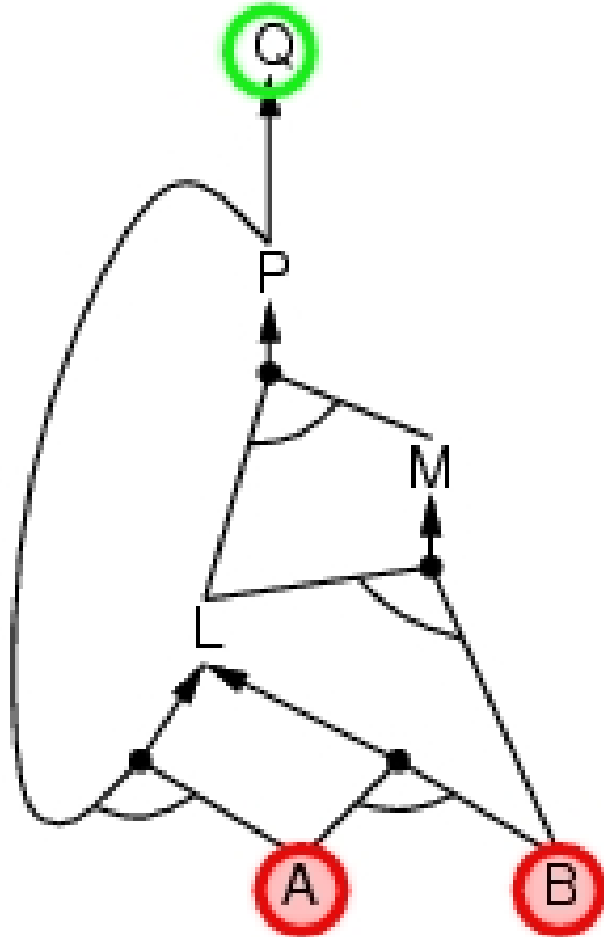
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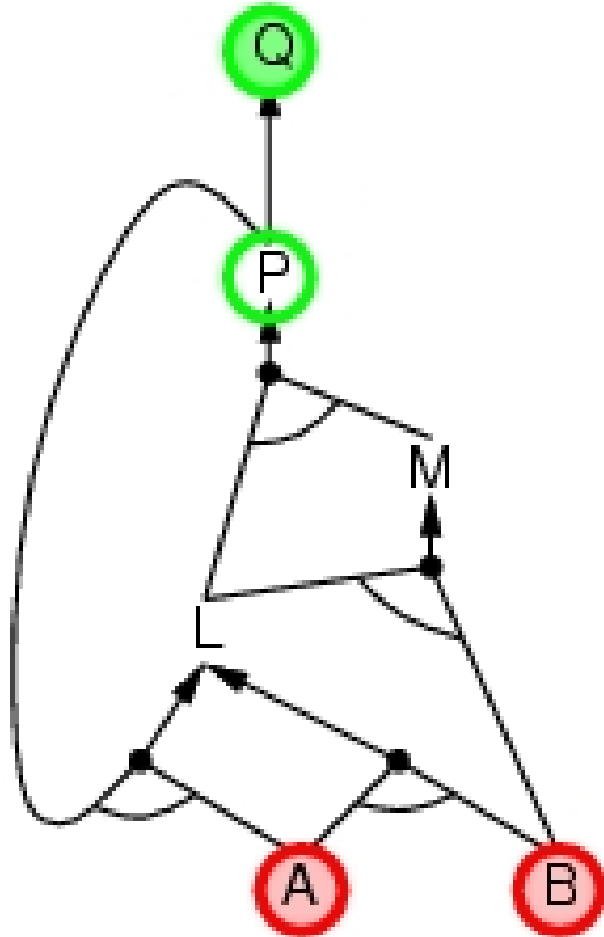
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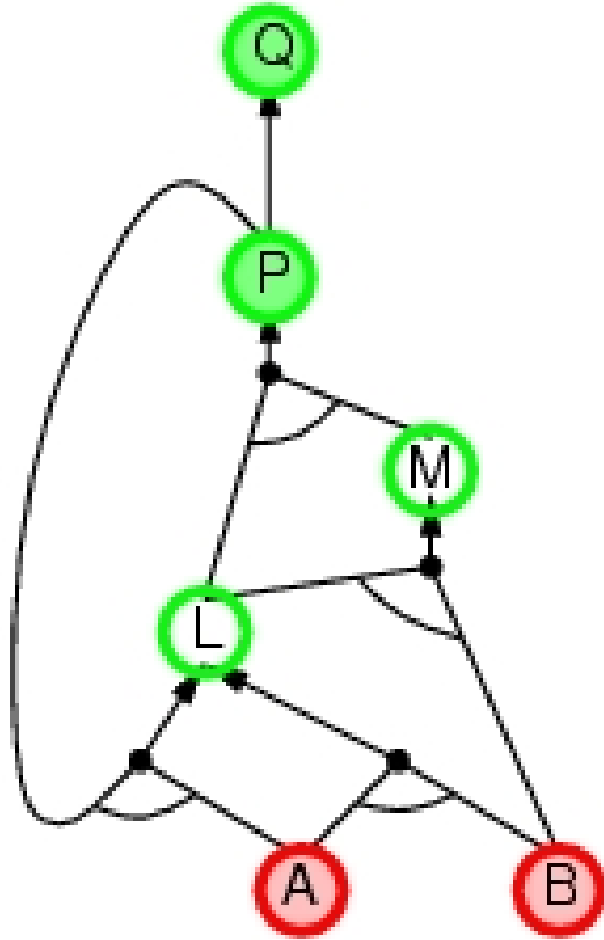
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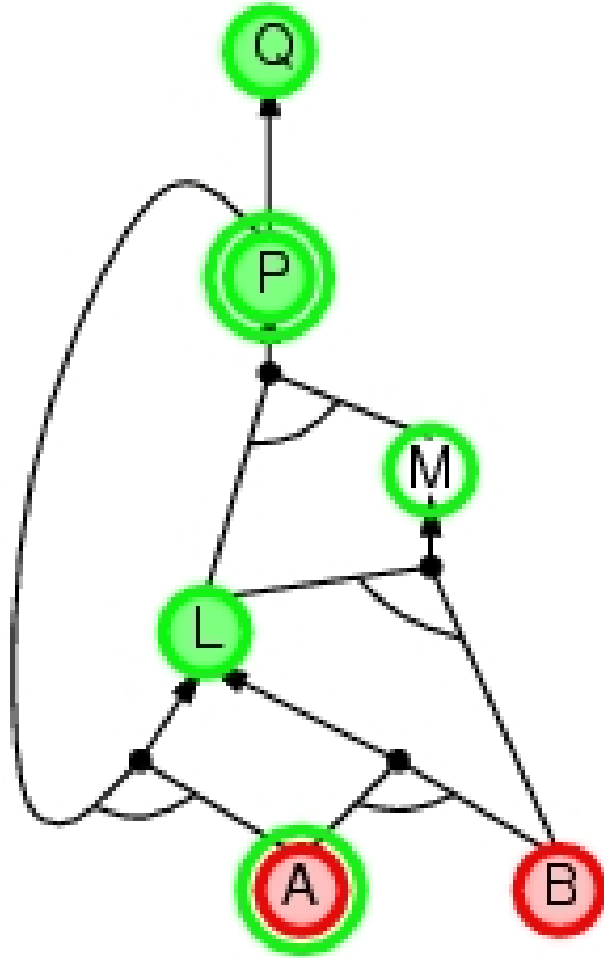
Backward chaining example



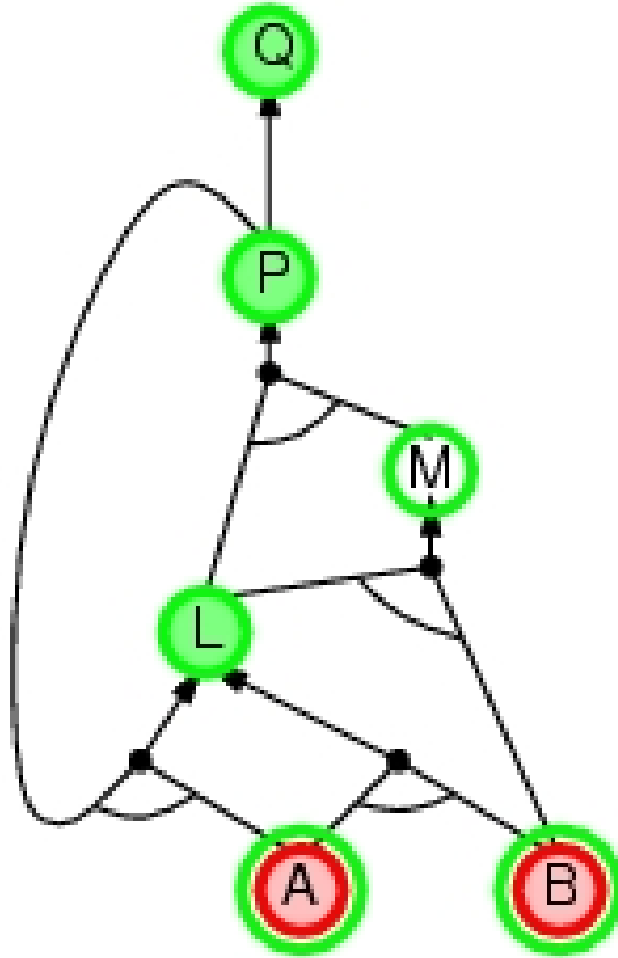
Backward chaining example



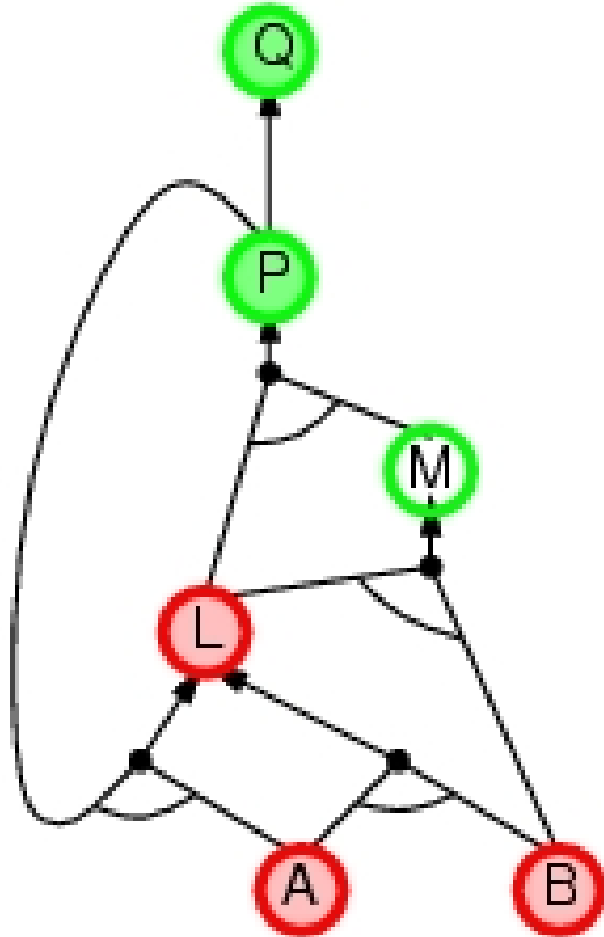
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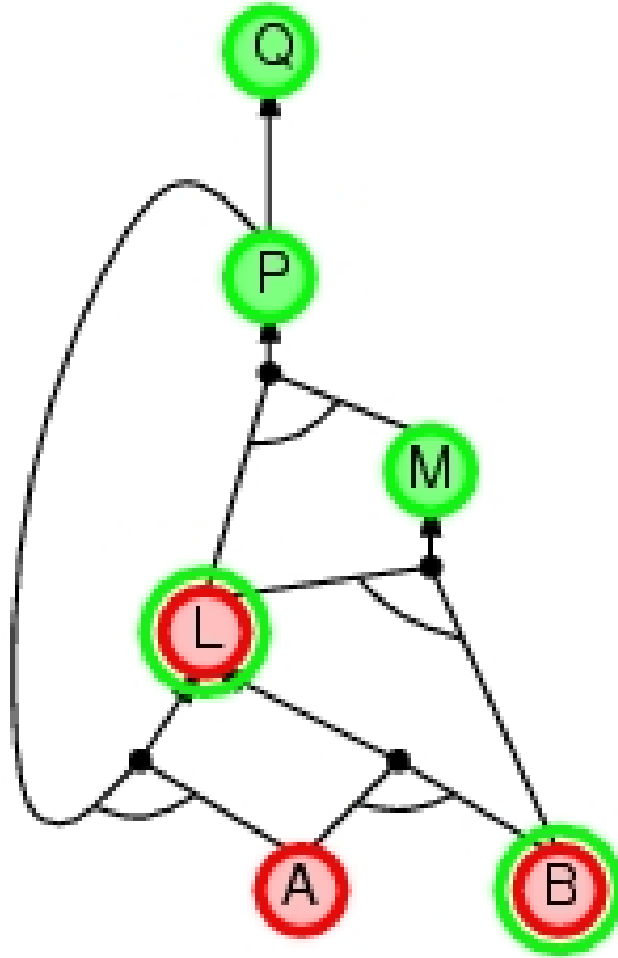
Backward chaining example



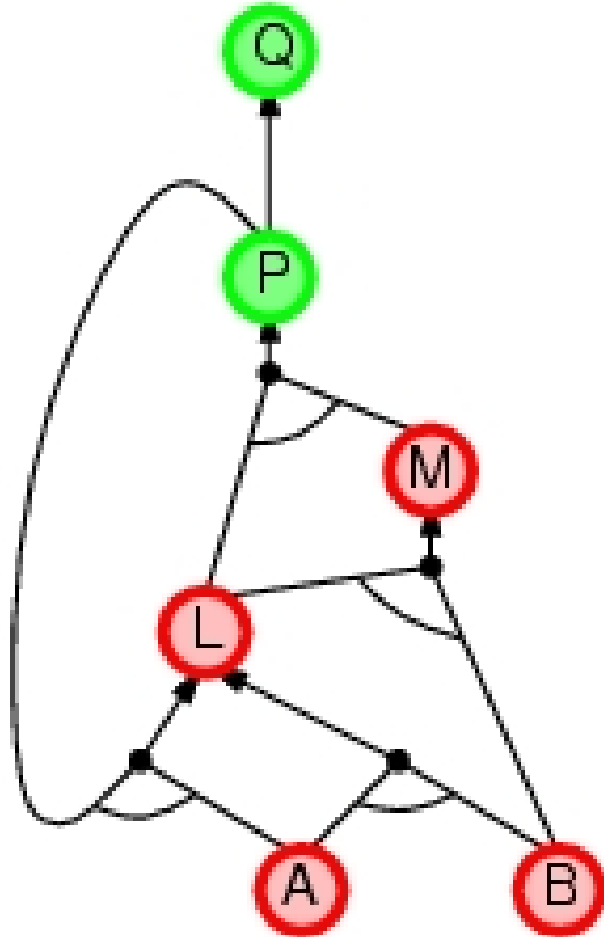
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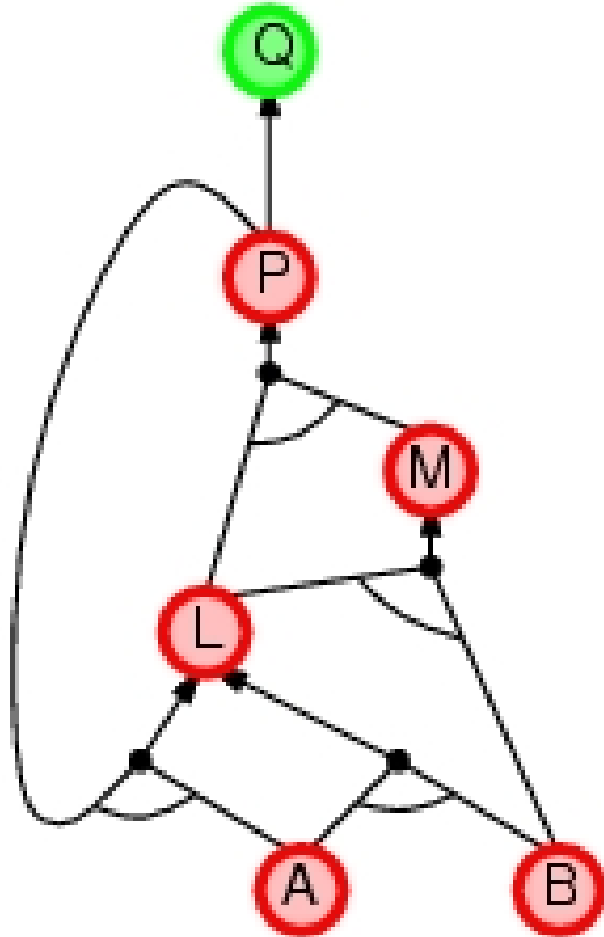
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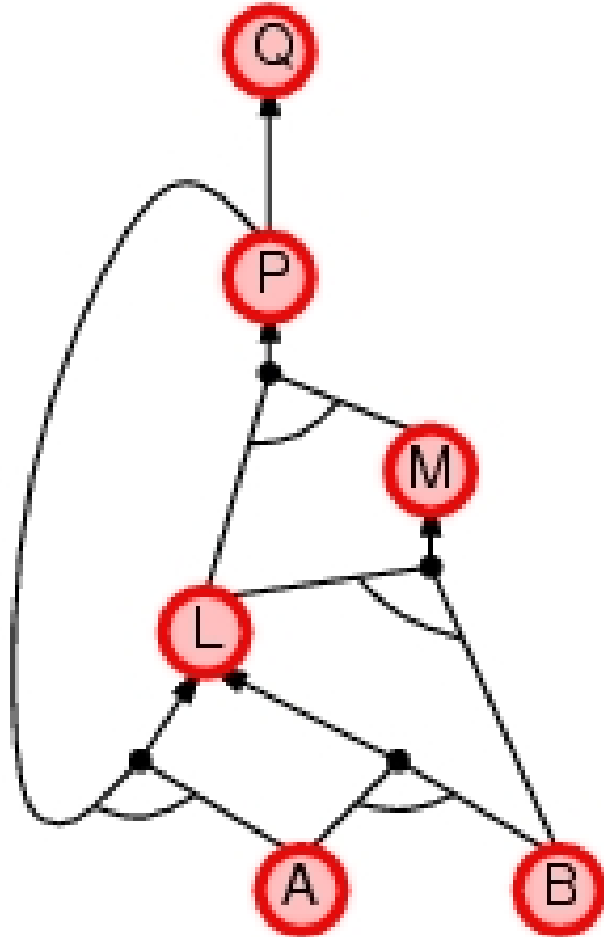
Backward chaining example



Backward chaining example



Backward chaining example



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Forward vs. backward chaining

Inference-based agents in the wumpus world

A wumpus-world agent using propositional logic:

$\neg P_{1,1}$

$\neg W_{1,1}$

$B_{x,y} \Leftrightarrow (P_{x,y+1} \vee P_{x,y-1} \vee P_{x+1,y} \vee P_{x-1,y})$

$S_{x,y} \Leftrightarrow (W_{x,y+1} \vee W_{x,y-1} \vee W_{x+1,y} \vee W_{x-1,y})$

$W_{1,1} \vee W_{1,2} \vee \dots \vee W_{4,4}$

$\neg W_{1,1} \vee \neg W_{1,2}$

$\neg W_{1,1} \vee \neg W_{1,3}$

...

$\Rightarrow$ 64 distinct proposition symbols, 155 sentences

```

function PL-WUMPUS-AGENT(percept) returns an action
  inputs: percept, a list, [stench, breeze, glitter]
  static: KB, initially containing the “physics” of the wumpus world
           x, y, orientation, the agent’s position (init. [1,1]) and orient. (init. right)
           visited, an array indicating which squares have been visited, initially false
           action, the agent’s most recent action, initially null
           plan, an action sequence, initially empty

  update x, y, orientation, visited based on action
  if stench then TELL(KB,  $S_{x,y}$ ) else TELL(KB,  $\neg S_{x,y}$ )
  if breeze then TELL(KB,  $B_{x,y}$ ) else TELL(KB,  $\neg B_{x,y}$ )
  if glitter then action  $\leftarrow$  grab
  else if plan is nonempty then action  $\leftarrow$  POP(plan)
  else if for some fringe square  $[i,j]$ , ASK(KB, ( $\neg P_{i,j} \wedge \neg W_{i,j}$ )) is true or
           for some fringe square  $[i,j]$ , ASK(KB, ( $P_{i,j} \vee W_{i,j}$ )) is false then do
           plan  $\leftarrow$  A*-GRAPH-SEARCH(ROUTE-PB( $[x,y]$ , orientation,  $[i,j]$ , visited))
           action  $\leftarrow$  POP(plan)
  else action  $\leftarrow$  a randomly chosen move
  return action

```

Summary